

Roll No.

TPE-401

B. TECH. (PETROLEUM ENGG.) (FOURTH SEMESTER) END SEMESTER EXAMINATION, June, 2023

PETROLEUM GEOLOGY

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) Describe all physical properties of oil and gas. (CO1)
(b) Differentiate between Paraffins, Cycloparaffins and Aromatics and also give *three-three* examples. (CO1)
(c) Explain in detail, how conversion of organic matter into hydrocarbon takes place. (CO1)
 2. (a) (i) Write a short note on Inorganic Theory of Hydrocarbons. (CO1, CO2, CO4)
(ii) Describe clastic and carbonate reservoir rocks.
(b) Explain the concept of Migration of Hydrocarbon and Also describe Primary and Secondary Migration. (CO1, CO2, CO4)
(c) Explain the types of information provided by Rock Eval pyrolysis testing of potential petroleum source rocks. (CO1, CO2, CO4)

P. T. O.

(2)

3. (a) Describe the characteristics of source rock types and explain depositional environments in which they are typically deposited. (CO1, CO4, CO5)
- (b) Discuss in detail various types of porosity found in carbonate and clastic sedimentary rock. (CO1, CO4, CO5)
- (c) Write short notes on the following : (CO1, CO, CO5)
- (i) Volumetric estimation
- (ii) Production history
4. (a) Describe source rock and also explain effective, potential and possible source rock. (CO4)
- (b) Explain source rock evaluation and also describe source rock evaluation in terms of quantity, quality and thermal maturation. (CO4)
- (c) What do you understand by TOC ? Explain. Also write why TOC and Vitrinite Reflectance is an important parameter. (CO4)
5. (a) Describe Hydrocarbon sources and Hydrocarbon reserve. (CO1, CO2, CO5)
- (b) Describe traps. What are the different types of reservoir traps ? With the help of a neat sketch, explain structural traps. (CO1, CO2, CO5)
- (c) (i) Write a short note on organic theory of Hydrocarbons. (CO1, CO2, CO5)
- (ii) Explain fire point, pour point and dew point.

Roll No.

TPE-402

**B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023
RESERVOIR ENGINEERING—I
Time : Three Hours
Maximum Marks : 100**

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are *twenty*.
(iv) Each sub-question carries 10 marks.

1. (a) An oil reservoir exists at its bubble-point pressure of 3000 psia and temperature of 160°F. The oil has an API gravity of 42° and gas-oil ratio of 600 scf/STB. The specific gravity of the solution gas is 0.65.

The following additional data are also available : (CO₂, CO₃)

Reservoir area 640 acres

Average thickness = 10 ft

Connate water saturation 0.25

Effective porosity = 15%

Calculate the initial oil in place in STB.

P. T. O.

- (b) (i) Given the following permeability data from a core analysis report, calculate the average permeability of the reservoir. (CO₂, CO₃)

Depth (ft)	Permeability, md
3998-02	200
4002-04	130
4004-06	170
4006-08	180
4008-10	140

- (ii) Given the following core data, calculate the geometric average permeability :

Sample	h_i , ft	k_i , md
1	1.0	10
2	1.0	30
3	0.5	100
4	1.5	40
5	20	80
6	1.5	70
7	1.0	15
8	1.0	50
9	1.5	35
10	0.5	20

- (b) (i) Given the following permeability data from a core analysis report, calculate the average permeability of the reservoir. (CO₂, CO₃)

Depth (ft)	Permeability, md
3998-02	200
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8	1.0	50
9	1.5	35
10	0.5	20

- (c) An incompressible fluid flows in a linear porous media with the following properties : (CO2, CO3)

$$L = 2000 \text{ ft}$$

$$h = 20'$$

$$\text{Width} = 300'$$

$$k = 100 \text{ md}$$

$$\text{Porosity} = 15\%$$

$$\mu = 2 \text{ cp}$$

$$p_1 = 2000 \text{ psi}$$

$$p_2 = 1990 \text{ psis}$$

Calculate:

- (i) Flow rate in bbl/day
 - (ii) Apparent fluid velocity in ft/day
 - (iii) Actual fluid velocity in ft/day
2. (a) Differentiate between retrograde gas condensate reservoir, wet gas reservoir and near critical gas condensate reservoirs. (CO1, CO2)
- (b) Explain all physical properties of crude oil. (CO2, CO3)
- (c) Define the following : (CO2, CO3)
- (i) Total porosity
 - (ii) Effective porosity
 - (iii) Microporosity
 - (iv) Macroporosity

- (v) Mesoporosity
 - (vi) Texture
 - (vii) Framework grains
 - (viii) Matrix
 - (ix) Sweet crude
 - (x) Sour crude
3. (a) Explain the concept of Darcy's law and describe radial, linear, spherical and hemispherical flow. (CO2, CO3)
- (b) Explain the concept that how fluid will flow through fractures, is it good for reservoir or not. Give reason for your answer. Differentiate between Uniform and Non-uniform flow, Steady and Unsteady flow. (CO2, CO3)
- (c) With the help of a table, explain Tissot classification. (CO2, CO3)
4. (a) Describe the concept of rock and fluid compressibility with types of compressibility. (CO4)
- (b) With the help of a diagram, explain the concept of wettability and capillary pressure. (CO4)
- (c) (i) Calculate the pressure difference, i.e., capillary pressure and capillary rise in an oil-water system from the following data : (CO4)
- Contact angle = 30°
- Water density = 1.0 gm/cm^3
- Oil density = 0.75 gm/cm^3
- $r = 10^{-4} \text{ cm}$
- Oil water surface tension = 25 dynes/cm

- (ii) A brine is used to measure the absolute permeability of a core plug. The rock sample is 4 cm long and 3 cm² in cross-section. The brine has a viscosity of 1.0 cp and is flowing a constant rate of 0.5 cm³/sec under a 2.0 atm. pressure differential. Calculate the absolute permeability.
5. (a) With the help of a diagram, explain the concept of water and gas conning. (CO5)
- (b) Differentiate between combination and gravity drive. (CO5)
- (c) Explain the concept of solution gas drive and gas cap drive with percentage of their recovery. (CO5)

Roll No.

TPE-403

**B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)**

END SEMESTER EXAMINATION, June, 2023

HEAT AND MASS TRANSFER

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) How the conductivities varies with respect to temperature in solid metals, liquid and gases ? Write the exceptional material which shows adverse properties with the temperature.

Solve: Two slabs each 260 mm thick have thermal conductivities of $14.5 \text{ W/m}^\circ\text{C}$ and $220 \text{ W/m}^\circ\text{C}$. If the temperature of the face of hot surface is 240°C and the outside surface of the other slab is 40°C , determine the heat flow rate per unit area through the composite system and the intermediate temperature.

(CO1, CO2)

P. T. O.

- (b) Define insulation and hence critical radius of insulation ? What are the applications of insulation ? Derive an expression for Critical radius of insulation for cylindrical pipe. What is the ratio of critical radius of insulation of sphere and cylinder ? (CO1, CO2)
- (c) Define heat transfer coefficient and hence the overall heat transfer coefficient ? Derive an expression for the overall heat transfer coefficient considering the convective and conductive heat transfer over the solid metal boundary. (CO3)
2. (a) Discuss free convection from a flat surface for a vertical plate, hot or cold suggested by Churchill and Chu. Is this equation applicable for an inclined plane if yes then what should be the minimum angle of inclination ? If No why ? (CO3, CO4)
- (b) Air flows over a flat surface, 2 m in length oriented in the direction of flow and of sufficient breadth, maintained at 150°C . The pressure is 1 atm and the bulk air temperature is 30°C . If the air velocity is 12 m/s, determine (i) the local heat transfer coefficient as a function of longitudinal position, (ii) the average heat transfer coefficient and (iii) the rate of heat loss from the surface.

The relevant physical properties of air are taken as :

$$\rho = 0.962 \text{ kg/m}^3, \mu = 2.131 \times 10^{-5} \text{ kg/m-s}, K = 0.031 \text{ W/m}^{\circ}\text{C}, C_p = 1.01 \text{ kJ/kg }^{\circ}\text{C}.$$

(CO1, CO4)

- (c) Define the following Numbers along with their physical significance and their mathematical expressions : (CO1, CO2)
- (i) Reynolds no.
 - (ii) Nusselt no.
 - (iii) Prandtl no.
 - (iv) Stanton no.
 - (v) Peclet no.

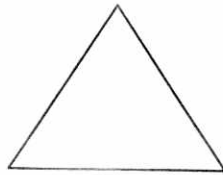
3. (a) Define Black and Gray Body. What are the laws and relations for radiation for surfaces on the basis of the concept of a black body ? Hence deduce the expression for Kirchhoff's Law (i. e., $\epsilon = \alpha$).

(CO1)

- (b) Define View factor. How many view factors involved, known and number of view factor that can be determined if we have 4-surface enclosure ? Calculate the View factor for the following configurations shown in the figure :

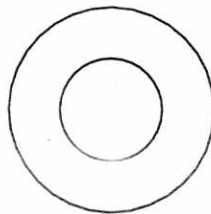
(CO1, CO2)

(i)



Equilateral triangle

(ii)



Sphere enclosed in a sphere

- (c) Derive an expression for the heat exchange between the nonblack bodies for infinite parallel planes mentioning the assumptions made for the analysis of heat exchange between two nonblack parallel surfaces. Specify the symbol used.
4. (a) Solve the following :
- (i) A refractory material which has $\epsilon = 0.4$ at 1500 K and $\epsilon = 0.43$ at 1420 K is exposed to black furnace walls at 1500 K. What is the rate of gain of radiation per m^2 area ?

P. T. O.

- (ii) Define radiation Shield ? Determine the radiant heat exchanger in W/m^2 between two large parallel steel plates of emissivities 0.8 and 0.5 held at a temperature of 1000 K and 500 K respectively, if a thin copper plate of emissivity 0.1 introduced as a radiation shield between the two plates. Use $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{ K}^4$.
- (b) Write the assumptions made during the design aspects of heat exchanger. Derive the relation of logarithmic mean temperature difference [LMTD] for counter flow heat exchanger. Specify the symbol used. (CO4, CO5)
- (c) Illustrate the following : (CO3, CO6)
- (i) LMTD
 - (ii) Effectiveness
 - (iii) Capacity ratio
 - (iv) NTU

And Hence Solve: A crude oil with a flowrate of 1000 kg/hr is to be cooled using water in a double-pipe counter-flow heat exchanger from a temperature of 80°C to 40°C . The water enters the exchanger at 20°C and leaves at 40°C . The specific heat capacities of the oil and the water at constant pressure are 2 kJ/kg K and 4.2 kJ/kg K , respectively. The overall heat transfer coefficient is $0.25 \text{ kW/m}^2 \text{ K}$. Neglecting the heat loss and using the log mean temperature difference (LMTD) method, the minimum heat exchanger area (m^2) required for the operation is (rounded off to two decimal places).

5. (a) Illustrate the modes of Mass Transfer. Define the mass transfer terms with the mathematical expression of : (CO4, CO6)
- (i) Concentrations
 - (ii) Velocities
 - (iii) Fluxes
- (b) Define Fick's law. What is mass transfer coefficient ? Derive general mass diffusion equation in a stationary media. (CO1, CO5)
- (c) Define mass transfer by convection. Define diffusional resistance and hence. Derive the expression for steady state diffusion through a plane membrane. (CO3, CO4)

Roll No.

TMA-403

**B. TECH. (PETROLEUM ENGG.) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

NUMERICAL AND GEOSTASTICAL METHODS

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) An analysis of production rejects resulted in the following figures :

No. of rejects per operator	No. of operators
21—25	5
26—30	15
31—35	28
36—40	42
41—45	15
46—50	12
51—55	3

Calculate the coefficient of skewness and kurtosis also comment on the results.

(CO1)

P. T. O.

- (b) If the mean of the following frequency distribution is 39.25, find the value of x : (CO1)

Daily wages in Rupees	Number of Persons
25	10
30	x
35	13
50	8
60	5
75	4

- (c) In the following table x is a discrete random variable and $p(x)$ is the probability density given as follows : (CO1)

x_i	$p(x_i)$
0	0.02
1	0.15
2	0.22
3	0.26
4	0.17
5	0.14
6	0.04

Find the standard deviation.

2. (a) In a nutritional study, 13 children were given a usual diet plus vitamins A and D tablets. While the second comparable group of 12 children was taking the usual diet. After 12 months, the gain in weight in pounds was noted as given in the table. Can you say that vitamins A and D were responsible for this difference ? (CO2)

Gain in weight in pounds

Vitamin A	Vitamin B
5	1
3	3
4	2
3	4
2	2
6	1
3	3
2	4
3	3
6	2
7	2
5	3
3	—

(Given : Tabulated value for 23 degrees of freedom at 5% level of significance is 1.71)

- (b) A company gives on the job training to its salesman which is followed by a test. It is considering whether it should terminate the services of any salesman who does not do well in the test. The following data give the test scores and sales made by nine salesman during the last one year :

Test Scores	Sales (in Thousands)
14	31
19	36
24	48
21	37
26	50
22	45
15	33
20	41
19	39

Compute the coefficient of correlation between test scores and sales. Does it indicate that the termination of the services of sales man with low test scores is justified ? (CO2)

- (c) The following are the values of two variables, diastolic blood pressure (y) and time in minutes (x). Diastolic blood pressure (y) was recorded at

various time periods (x) on individuals. Find the regression line of y on x on y also find the coefficient of correlation : (CO2)

Times periods (x)	Diastolic blood pressure (y)
0	72
5	67
10	70
15	65
20	66

3. (a) Two research laboratories have independently produced drugs that provide relief to arthritis patients. The first drug was tested on a group of 100 arthritis patients and produced an average of 8.5 hours of relief with a standard deviation of 2 hours. The second drug was tested on 75 patients, producing an average of 7.8 hours of relief with a standard deviation of 1.5 hour. At a significance level of one percent, does the first drug provide a significantly longer period of relief. (CO3 and CO4)
- (b) A sample of 100 battery cells is tested to find length of life with mean 10 hours and standard deviation 4 hours. Assuming that the data is normally distributed, what percentage of battery cells are expected to have life : (CO3 and CO4)
- (i) more than 13 hours
 - (ii) less than 5 hours

- (c) A certain drug was administered to 500 people out of a total 800 included in the sample to test its efficacy against typhoid. The results are given below : (CO3 and CO4)

	Typhoid	No Typhoid
Drug	200	300
No Drug	280	20

On the basis of these data, can it be concluded that the drug is effective in Preventing typhoid. (**Given** : Tabulated value for 1 degree of freedom at 5% level of significance is 3.841).

4. (a) Using Least square method to fit a curve $y = ax^2 + bx + c$ for the following data : (CO4)

x	y
1	10
3	5
5	20
10	25
15	22

- (b) The following table gives the number of good and bad parts produced by each of three shifts in a factory : (CO4)

Shift	Good	Bad
Day	900	130
Evening	700	170
Night	400	200

Is there any association between the shift and the equality of parts produced ? Chi-Square tabulate value is 5.991.

- (c) Calculate the standard deviation for the following frequency distribution :
(CO4)

x	f
2	25
3	37
4	44
5	59
9	68
10	43
12	31
13	32
15	12

5. (a) Define the linear regression, and analysis of variance for multiple regression with the help of an example. (CO5)

- (b) A data set containing n ($= 10$) independent measurements (x_i, y_i) is to be fitted by a simple linear regression model. The least square estimates of regression coefficients are obtained and the regression estimate is

$$\text{given by } \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x, \hat{\beta}_0 = \frac{\sum_{i=1}^n y_i}{n} - \hat{\beta}_1 \frac{\sum_{i=1}^n x_i}{n} \text{ and } \hat{\beta}_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)},$$

Where, $C(x, y)$ is the sample covariance and $\text{Var}(x)$ is the sample variance.

The values are given below :

$$\sum_{i=1}^{10} x_i = 25, \sum_{i=1}^{10} y_i = 37, \sum_{i=1}^{10} x_i^2 = 108, \sum_{i=1}^{10} y_i^2 = 155$$

$$\text{and } \sum_{i=1}^{10} x_i y_i = 120$$

For the given data set, calculate the unbiased variance for the error $(y_i - \bar{y}_i)$ (rounded off to two decimal places). (CO5)

- (c) Explain normal distribution. How is it important in geological sequences ? Write the formula for normal distribution. Draw a figure of normal distribution with variance. (CO5)

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TPE-404

**B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023
APPLIED SEDIMENTOLOGY**

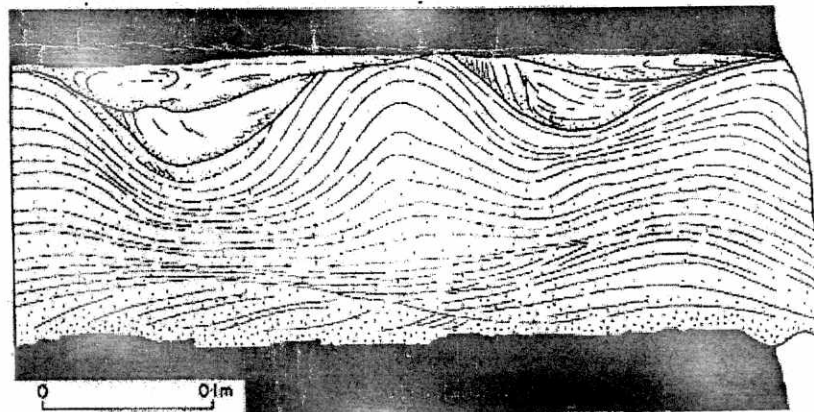
Time : Three Hours

Maximum Marks : 100

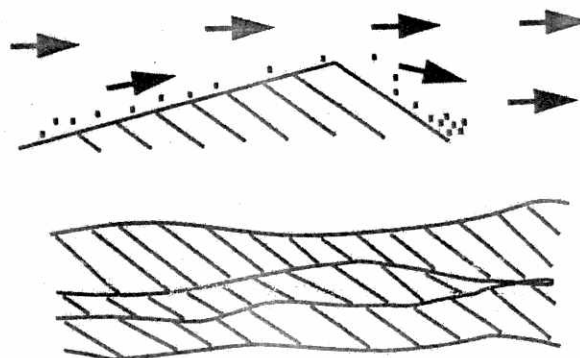
- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are *twenty*.
(iv) Each sub-question carries 10 marks.
1. (a) (i) Determine the flow of fluid having a relative density of 100 kg/m^3 , the viscosity of 0.5 Ns/m^2 with a velocity of 5 m/s through a pipe of 0.2 m . (CO2)
(ii) Calculate the Reynolds number, Re , for oil flow in a circular pipe. The diameter of the pipe is 60 mm , the density of the oil is 910 kg/m^3 , the volumetric oil flow rate is 60 L/min , and the dynamic viscosity of the oil is 50 mPa s .

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- (b) (i) Determine the Froude number for a fluid flow in an open channel having area of cross-section, 1 m^2 and width 0.5 m . The flow velocity is 1 m/s . Use acceleration due to gravity as 9.81 m/s^2 . (CO2)
- (ii) Suppose the water is flowing through a pipe with a diameter of 3.5 cm . The velocity with which the water is flowing is 1.5 m/s . If the density of water is 1000 kg/m^3 and the coefficient of viscosity of water is $9 \times 10^{-4} \text{ Pa.s}$. Find the Reynolds number to determine if the flow of water is streamlined or turbulent.
- (c) In brief describe mode of sediment transport. (CO2)
2. (a) Identify the sedimentary structure and describe that sedimentary structure and also explain in which environment they can be seen. (CO3)



- (b) Identify the sedimentary structure and describe that sedimentary structure and also explain in which environment they can be seen. (CO3)



(3)

- (c) Differentiate Pre-depositional Interbed and Syn-depositional intrabed. (CO3)
3. (a) What is the significance of cross beds and ripples. (CO2, CO3)
- (b) What is porosity ? Discuss all the parameters on which porosity of rocks depends. (CO2, CO3)
- (c) What is the clastic sedimentary rock ? Write the name of two clastic sedimentary rocks. Give the classification of clastic sedimentary rocks. (CO2, CO3)
4. (a) Explain Sandstone and mudstone with their classification. (CO3, CO5)
- (b) With the help of a diagram describe massive and graded bedding. (CO3, CO5)
- (c) Describe the tectonic control of sedimentation. (CO3, CO5)
5. (a) Describe Physical & Chemical Parameters of depositional Environment. (CO2, CO4)
- (b) With the help of table explain Wentworth grain size classification. (CO2, CO4)
- (c) What are the differences between physical weathering and chemical weathering ? What are the main agents responsible for physical weathering and chemical weathering. (CO2, CO4)

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TPE-405

**B. TECH. (PETROLEUM ENGINEERING) (FOURTH
SEMESTER)**

END SEMESTER EXAMINATION, June, 2023

PETROLEUM DRILLING ENGINEERING-I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the component and functions of rotating system. (CO1)
- (b) A drillstring is composed of 7000 ft of 5-in, 19.5-lbm/ft drillpipe and 500 ft of 8-in OD by 2.75-in ID drill collars when drilling 9.875-in borehole. Assuming that the borehole remains in gauge compute the number of pump cycles required to circulate mud from the surface to the bit and from the bottom of the hole to the surface if the pump factor is 0.1781 bbl/cycle. (CO1)
- (c) What is the purpose of well control and monitoring system ? Describe the primary and secondary signals of kicks. (CO1)

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2. (a) What is drilling fluid ? What are the different types of drilling fluid ?
List the main functions of drilling fluid. (CO3)
- (b) While drilling an $8\frac{1}{2}$ in hole at 17523 ft (TVD) with a mud density of 11 ppg, the well encountered a big limestone cavern. Therefore, there was mud loss. Drilling was stopped, and the annulus was filled with 58 bbls of 8.4 ppg water until the well was stabilized. Calculate the formation pressure and the density that should be used to drill through the zone. The previous 9 5/8 in casing was set at 15500 ft.; the drilling consists of 900 ft of 6 in drill collar and 5 in drillpipe. Use the capacity of the casing annulus against the drillpipe to be 0.05149 bbl/ft. (CO2)
- (c) Describe yield point and gel strength of drilling fluid. Discuss its significance. Calculate how many sacks of barite are required to increase the density of an 800 barrel mud system from 12.7 lb/gal to 14.5 lb/gal. (CO3)
3. (a) What are the purpose different casing ? What is the difference between casing and liner ? What are the functions of cementing ? What are primary, liner, squeeze and plug back cementing ? Explain. (CO4)
- (b) What is key-seating phenomena during cementing operations in and oil well ? How can the key-seating be prevented ? Describe the remedies that can be. (CO4)
- (c) Discuss the classification of cement as per API. (CO4)
4. (a) What is hydrostatic pressure ? Distinguish between normal and abnormal pressure. Discuss the causes of abnormal pore pressure. (CO5)

(3)

- (b) Determine the slip velocity and transport velocity for the following well : (CO5)

Depth = 9000 ft

Hole diameter = 12.25-in

Drill pipe = 5-in/4.276-in

Flow rate = 500 gpm

Mud density = 10 ppg

Density of rock = 21 ppg

Average particle size = 0.28-in

Viscometer readings : $\theta_{600} = 90$

$\theta_{300} = 50$

- (c) List and describe the problems when designing cement slurry for shallow formations in deep water. (CO5)
5. (a) List the *two* main types of bits in use today. Also list two sub-classifications of each basic bit type and discuss the conditions considered ideal for the application of each sub-classification given. (CO6)
- (b) List *seven* rules of thumb of bit selection. What is the best basis of comparison when trying to choose between two different bit types ? (CO6)
- (c) Discuss the factors affecting penetration of a drill bit. (CO6)

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TPE-413

**B. TECH. (PETROLEUM ENGG.) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

MUD LOGGING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) How is the mud logging is useful in Formation Evaluation ? Explain lag-up, lag-down and complete cycle in detail. (CO1)
- (b) What is Lag time and Lag strokes ? Why is it important ? How is it calculated ? (CO2)
- (c) What are the various sensors used for recording mud logging data ? What are the various mud logging parameters recorded during data acquisition ? (CO3)
2. (a) What is mud log ? How the various components of mud log to be put in various tracks for interpretation ? Draw a diagram. (CO3)
- (b) What are the uses of drilling cutting analysis during mud logging ? What is shale shaker and why is it required ? (CO3)

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- (c) Explain Coring. Why is it required ? What are the parameters measured from coring ? (CO4)
3. (a) How many type of coring are taken during drilling process ? Explain each in detail. (CO4)
- (b) What are the various constituents of sieved cuttings sample ? Why these are important ? (CO4)
- (c) Write the core handling process in detail. What are the precaution required for core handling ? (CO4)
4. (a) Write the fundamental gas properties in detail. What is the difference between conventional gas reservoirs and CBM gas reservoir ? (CO4)
- (b) Explain gas hydrates. How are they formed ? What are the essential condition for gas hydrate ? (CO5)
- (c) Explain Hydrostatic pressure, Pore pressure, abnormal pressure, over-pressure, Under-pressure in detail. Why are they important in oil industry ? (CO5)
5. (a) Explain Viscosity, Reynold's Number. Why water changes its flow from laminar to turbulent when is heated ? (CO5)
- (b) How many type of drilling fluid are uses during drilling operation ? What are the condition for using different drilling fluids in drilling process ? (CO5)
- (c) Write Drilling Mud Properties. What are the various contaminants in Drilling Fluids ? Explain each in detail. What are its side effect ? (CO5)